

Table of Contents

CHAPTER 1: Introduction.....	1-2
1.2 Objectives	1
1.3 Scope.....	2
CHAPTER 2: System Design	3
CHAPTER 3: Application Features	6
CHAPTER 4: Technical Implementation	9
CHAPTER 5: Performance Optimization	13
CHAPTER 6: AI-Powered Anomaly Detection.....	14
CHAPTER 7: Testing and Quality Assurance.....	16
CHAPTER 8: Deployment and Maintenance.....	19
CHAPTER 9: Conclusion and Future Scope.....	21
References	23
Acknowledgement.....	24

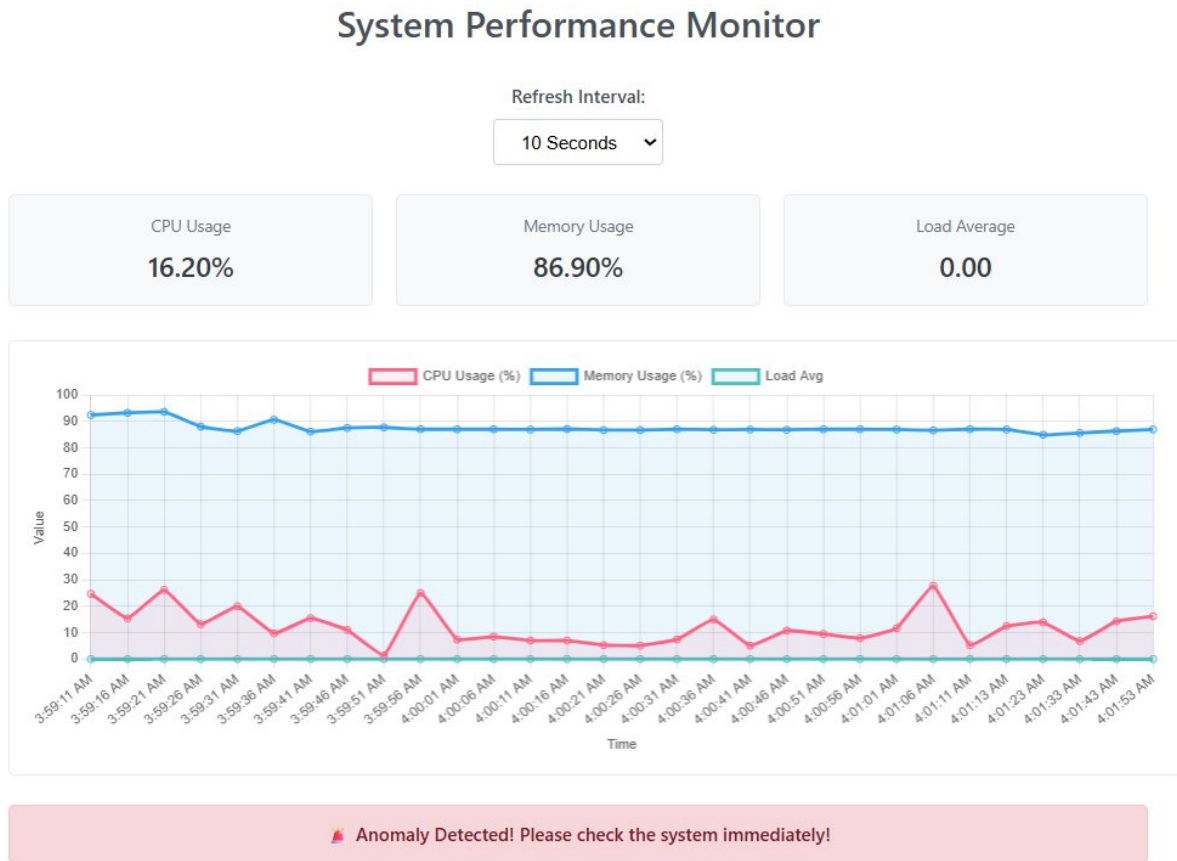


Figure. App Screenshot

```
(venv) D:\performance-monitor-main>python run.py
2025-06-28 03:58:30,303 - app - INFO - Flask application factory initialized.
2025-06-28 03:58:42,865 - app - INFO - Starting development server...
* Serving Flask app 'app'
* Debug mode: off
2025-06-28 03:58:42,897 - werkzeug - INFO - WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://192.168.0.102:5000
2025-06-28 03:58:42,897 - werkzeug - INFO - Press CTRL+C to quit
2025-06-28 03:59:11,111 - werkzeug - INFO - 127.0.0.1 - - [28/Jun/2025 03:59:11] "GET / HTTP/1.1" 302 -
2025-06-28 03:59:11,207 - werkzeug - INFO - 127.0.0.1 - - [28/Jun/2025 03:59:11] "GET /dashboard HTTP/1.1" 200 -
2025-06-28 03:59:11,435 - werkzeug - INFO - 127.0.0.1 - - [28/Jun/2025 03:59:11] "GET /static/styles.css HTTP/1.1" 200 -
2025-06-28 03:59:11,476 - werkzeug - INFO - 127.0.0.1 - - [28/Jun/2025 03:59:11] "GET /static/script.js HTTP/1.1" 200 -
2025-06-28 03:59:11,712 - werkzeug - INFO - 127.0.0.1 - - [28/Jun/2025 03:59:11] "GET /favicon.ico HTTP/1.1" 404 -
2025-06-28 03:59:11,772 - app - INFO - Loading model from models\anomaly_model.pkl
2025-06-28 03:59:11,849 - app - WARNING - Anomaly detected: {'cpu_usage': 24.7, 'memory_usage': 92.4, 'load_avg': 0.0}
2025-06-28 03:59:11,849 - app - WARNING - Email credentials not configured. Skipping alert.
2025-06-28 03:59:11,849 - werkzeug - INFO - 127.0.0.1 - - [28/Jun/2025 03:59:11] "GET /metrics HTTP/1.1" 200 -
```

Figure. App Backend Screenshot

ABSTRACT

During my 5-month internship at **BEL Smart City Division**, I independently designed and developed a real-time **Performance Monitoring System with AI-Powered Anomaly Detection** to support infrastructure-level observability needs. This intelligent platform tracks key system metrics—including CPU usage, memory consumption, and load averages—and proactively identifies anomalies using a trained machine learning model.

The backend, built using the lightweight **Flask** web framework, exposes RESTful API endpoints and supports robust logging. The frontend uses **HTML**, **CSS**, and **JavaScript** with **Chart.js** to render real-time visualizations. The system is containerized using **Docker** and deployed on **Render**, enabling seamless, version-controlled deployments via Git.

A key innovation is the use of the **Isolation Forest** algorithm to detect abnormal patterns. Upon identifying a potential system failure or threshold breach, the system sends real-time email alerts to designated administrators, enabling proactive incident handling.

Developed from scratch within a production-facing smart city context, this modular and scalable solution demonstrates how AI and modern DevOps tools can be effectively integrated to build intelligent monitoring solutions tailored for critical infrastructure environments.